



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

Name: Aman S. Choudhary

PRN: 25070122033

Div: CSE A1

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ASSIGNMENT No :2

TITLE :Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Assignment 2: Methods and Static Members

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Outcome

By the end of this exercise, students will be able to understand and implement default, parameterized, and copy constructors in Java, and apply them to initialize and manage object data.

Conclusion

Thus, we have successfully studied and implemented default, parameterized, and copy constructors in Java. The Student class demonstrated how default and parameterized constructors initialize object data differently, while the Mobile Phone Inventory System demonstrated how a copy constructor can be used to duplicate existing object records. The programs were executed successfully, and the outputs verified the correct working of all three constructor types.

Exercise

1. **Create a Student class using default and parameterized constructors to initialize student name and roll number.**

```
J JavaTest.java > ...
1  class Student {
2      String name;
3      int rollNo;
4
5      // Default constructor
6      Student() {
7          name = "Unknown";
8          rollNo = 0;
9      }
10
11     // Parameterized constructor
12     Student(String n, int r) {
13         name = n;
14         rollNo = r;
15     }
16
17     void display() {
18         System.out.println("Name: " + name + ", Roll No: " + rollNo);
19     }
20 }
21
22 public class JavaTest {
23     Run | Debug
24     public static void main(String[] args) {
25         Student s1 = new Student();           // uses default constructor
26         Student s2 = new Student(n: "Aditi", r: 21); // uses parameterized constructor
27
28         s1.display();
29         s2.display();
30     }
31 }
32
33 Name: Unknown, Roll No: 0
34 Name: Aditi, Roll No: 21
```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

```

1  class MobilePhone {
2      String brand;
3      double price;
4
5      // Default constructor
6      MobilePhone() {
7          brand = "Not Set";
8          price = 0.0;
9      }
10     // Parameterized constructor
11     MobilePhone(String b, double p) {
12         brand = b;
13         price = p;
14     }
15     // Copy constructor
16     MobilePhone(MobilePhone m) {
17         brand = m.brand;
18         price = m.price;
19     }
20     void display() {
21         System.out.println("Brand: " + brand + ", Price: " + price);
22     }
23 }
24 public class MobileInventoryDemo {
25     Run | Debug
26     public static void main(String[] args) {
27         MobilePhone m1 = new MobilePhone();           // default
28         MobilePhone m2 = new MobilePhone(b: "Samsung", p: 25000); // parameterized
29         MobilePhone m3 = new MobilePhone(m2);         // copy of m2
30
31         m1.display();
32         m2.display();
33         m3.display();
34     }
35 }

```

```

Brand: Not Set, Price: 0.0
Brand: Samsung, Price: 25000.0
Brand: Samsung, Price: 25000.0

```
